

The cold case

Teacher guide and worked answers

Use evidence to explain thermal conduction and select insulation for a cold container.

40 minutes. Use Lesson.html for interactive teaching or Lesson.pptx for editable slides. Slides.pdf is the static alternative. Select one pupil response route and print the shared evidence it requires. Detailed answers below are for staff.

Scheme of work

[_passsg/inputs/GROW SOW 2026-27.xlsx](#) | GROW Weekly - Spring | B30:C30

Investigate thermal and electrical conductors and insulators.

Directly develops the stated science outcome. Original secondary-age exemplar activities; diagnostic-led support and stretch follow the SoW. No claim of qualification approval.

Prior learning

Identify common materials and compare a measured temperature.

Use higher/lower and read whole-number values in a table.

Success evidence

Show the direction of energy transfer between warmer and cooler objects.

Explain why a thermal conductor transfers energy more readily than an insulator.

Choose an insulation jacket using two numbers and identify one test limitation.

Equipment

Printed evidence cards and one selected response route per pupil; pencils; optional calculator.

Optional teacher demonstration: smooth metal and wooden blocks kept in the room; contact thermometer.

No hot water, heater, ice handling or practical apparatus is needed for the complete lesson.

Preparation

Print the common evidence page for every pair and choose response pages from the retrieval check.

If using the optional touch comparison, leave both safe samples in the same room well in advance and measure them before teaching.

Keep the constructed data visible. These are teaching examples, not results collected by this class.

Practical care

For the optional demonstration, use smooth, clean samples without sharp edges or splinters.

Touch is optional; pupils may observe temperature readings instead. Do not touch heated or chilled objects.

Use the complete paper route if safe sample preparation or supervision is unavailable.

Ready to teach alternative

The default route uses supplied diagrams and constructed evidence. It teaches the complete objective without handling hot or cold materials.

Teaching sequence

Slide	Stage	Minutes	Focus
1	Opening	0-2	Can a jacket keep something cold?
2	Retrieval	2-5	What do we already know?
3	I do	5-13	Feeling is not a thermometer
4	I do	Within stage	Why can ice melt faster on metal?
5	I do	Within stage	Insulation changes the rate
6	We do	13-21	Evidence detectives: feeling versus measurement
7	We do	Within stage	Which bottle warmed least?
8	We do	Within stage	Improve the delivery claim
9	Hinge	21-24	A cold bottle wears an insulating jacket
10	You do	24-34	Your design brief
11	You do	Within stage	Make the evidence do the work
12	You do	Within stage	Check before you submit
13	Review	34-38	Compare and improve
14	Review	Within stage	Our evidence has a boundary
15	Exit	38-40	Three things to leave with

Times are a suggested 40-minute route. Slides marked Within stage share the preceding stage allocation. The optional HTML timer starts only when selected.

1 Opening Can a jacket keep something cold?

Introduce an ordinary cold-drink delivery, with no health or food-storage guarantee. The scientific decision concerns only temperature change over 20 minutes. Say: We can change our first idea when evidence improves it.

2 Retrieval What do we already know?

Answers: a suitable metal such as steel; 20 °C; thermometer. Accept everyday language, then supply the scientific word. If values are confused, locate 8 and 20 on an imagined number line. Do not teach relative temperature by touching unknown objects.

3 I do Feeling is not a thermometer

8 minutes shared across this and the next two I do slides. Think aloud: I notice the two thermometer readings match. So lower starting temperature cannot explain the different feeling. I compare how quickly energy leaves the warmer hand. Optional smooth room-temperature sample observation only. The about-32 °C hand is an illustrative value, not a body-temperature claim.

4 I do Why can ice melt faster on metal?

Shared I do time. Trace the arrows from tray towards ice. Conduction through metal helps supply energy for melting. Explain that energy also enters from air and radiation; the diagram isolates the contact route. Do not imply the tray stays at 20 °C throughout or that all metals behave identically.

5 I do Insulation changes the rate

Shared I do time. Model the cold case: 20 °C room is warmer than 8 °C contents, so energy transfers inwards. A foam jacket slows this. Trapped gas in foam reduces conduction and limits bulk air movement; do not claim conduction is the only route in a real bottle.

6 We do Evidence detectives: feeling versus measurement

8 minutes shared across three We do slides. Give pairs the common evidence page. Roles: evidence finder and claim editor, with silent individual alternative. D1 rejects metal must be colder. D2 is consistent with faster energy transfer through the metal set-up. It alone does not measure conductivity because dimensions, masses, contact and other routes affect melting.

7 We do Which bottle warmed least?

Shared We do time. Ask everyone to first point at the lowest final temperature. Compute rises together: $14-8=6$ °C, $12-8=4$ °C, $10-8=2$ °C. The foam jacket performed best among these tested jackets. Table concerns a single model comparison; repeat before a stronger conclusion.

8 We do Improve the delivery claim

Shared We do time. Model after pupil attempts: In this test, the foam jacket limited warming to 2 °C, compared with 6 °C without a jacket. It slowed energy transfer into the bottle. Repeat tests; record thickness and fit; test longer journeys. Accept equivalent bounded conclusions.

Reveal: This foam jacket slowed warming most in this 20-minute test.

9 Hinge A cold bottle wears an insulating jacket

Take simultaneous responses by pointing, cards or private writing. If A or C chosen, revisit the cold-container arrow before independent work. If D chosen, ask which value remains lower than the room temperature.

A. The jacket produces cold. - No. A passive jacket does not produce cold; it slows energy transfer.

B. Energy still transfers in, but more slowly. - Yes. The room is warmer, so net energy transfer is inwards; insulation slows it.

C. The jacket stops all transfer forever. - No. Insulation reduces the rate; temperatures can still move towards room temperature.

D. Energy transfers from cold bottle to warmer room. - No. The net transfer in this situation is from the warmer room to the cooler bottle.

10 You do Your design brief

10 minutes shared across the three You do slides. All routes assess the same objective. Select after the hinge, not from a fixed ability label. Provide page 1 with each route. Read prompts aloud or scribe verbatim where needed. Independent explanation can be drawn and dictated.

11 You do Make the evidence do the work

Shared You do time. Supported: 10 versus 14 °C and inward arrow suffice with a clear explanation.

Standard: show all rises. Stretch: compare a thick foam sleeve with a thin cotton sleeve fairly, and explain warm-container transfer. Prompt with questions, not the conclusion.

12 You do Check before you submit

Shared You do time. Teacher checks one explanation per pupil. For metal feels colder misconception, return to identical thermometer labels. For cold escapes, ask which object gives energy to which. Do not add new technical vocabulary during independent work.

13 Review Compare and improve

4 minutes shared across two Review slides. Offer private teacher sharing. Listen for foam jacket slowed warming; 2 °C rather than 6 °C; repeat under controlled conditions. Praise a specific revision of reasoning rather than speed or confidence.

14 Review Our evidence has a boundary

Shared Review time; light Lundy cycle. Invite a spoken, written or pointed proposal; teacher records one and explains which can be tested next, such as equal material at two thicknesses. Explain that listening led to a planned test, not merely collecting a vote. No obligation to disclose feelings.

15 Exit Three things to leave with

Answers: arrow room to bottle; slows energy transfer; 10 °C final or 2 °C rise, ideally compared with 14 °C or 6 °C unwrapped. Collect independent evidence; an adult may read and scribe without supplying concepts.

Answers and assessment

R1-R3

R1: steel/stainless steel or another suitable metal. R2: 20 °C. R3: a thermometer. Re-teach reading temperature values if needed.

W1 / C1

Reject C1. Both samples measured 20 °C. The warmer hand transfers energy to the metal faster, so feeling alone does not establish lower temperature.

W1 / C2

Keep the bounded claim C2: more ice melted in the metal set-up in the same time (9 g versus 3 g), consistent with faster net energy transfer into the ice in this set-up. It does not isolate conductivity alone; tray masses differed and other energy routes exist.

W1 / C3

Improve C3: the foam jacket slowed warming most in this test. It does not make cold and the test does not prove superiority under all conditions.

W2

Foam: final 10 °C and rise 2 °C, versus no jacket 14 °C and rise 6 °C. Useful checks include repeated trials; matched thickness if comparing material properties; matched jacket fit; a longer journey test.

Hinge A-D

A incorrect: a passive insulator makes no cold. B correct: net transfer is into the colder bottle and is slower with insulation. C incorrect: it does not stop transfer forever. D incorrect: direction is warmer room to cooler bottle. Use the cold-container diagram if A, C or D persist.

S1-S3

S1 room. S2 arrow from ROOM (20 °C, warmer) towards BOTTLE (8 °C, cooler). S3 foam at 10 °C.

S4-S6

S4 foam; 10; 14; slows. S5 the warmer hand loses energy faster to metal although both blocks begin at the same temperature. S6 repeat the test, to check whether the pattern happens again. Colour alone does not establish this performance.

N1-N2

N1 room to bottle. N2 none: $14-8=6$ °C; cotton: $12-8=4$ °C; foam: $10-8=2$ °C. Foam warmed 4 °C less than no jacket over the same interval.

N3-N5

N3 recommend this foam sleeve: it slowed energy transfer in from the warmer room, with only a 2 °C rise compared with 6 °C unwrapped. N4 both are 20 °C; metal conducts energy away from the warmer hand faster. N5 one set of results lacks repeat confidence; repeat each test. Thickness was not matched; match it to compare material properties. Check and report fit carefully too. Accept other specific, relevant control improvements.

T1-T2

T1 rises 6, 4 and 2 °C. Recommend foam for this brief; energy enters from the warmer room, more slowly with the jacket. T2 no: thickness changed as well as material. Compare equal-thickness, equal-coverage jackets with identical bottles, starting values and conditions; repeat. This tests those equal-thickness samples, not every possible design.

T3-T4

T3 arrow from 40 °C contents to 20 °C room; insulation slows energy loss so the contents cool more slowly. No physical hot-water method is requested. T4 the evidence compares full tray set-ups, not conductivity alone. Tray masses differed; use matching dimensions, control initial temperatures and contact, document masses and repeat. Even then, heat capacity and other transfer routes require care before attributing the effect solely to conductivity.

E1-E4

E1 room to bottle. E2 slows energy transfer by conduction; accept a correctly explained broader slows heat transfer. E3 foam 10 °C or 2 °C rise; stronger evidence compares 14 °C or 6 °C unwrapped. E4 a testable suggestion such as same material at two thicknesses, repeated trials, or a longer fixed duration. Reflection has no prescribed opinion; assess connection to evidence.

Assessment / re-teach

Secure objective: correct warmer-to-cooler arrow, explains an insulator slows transfer, and numerical recommendation bounded to this test. A correct choice without an explanation is emerging evidence. For cold moves or jackets make heat, narrate the two-temperature diagram, then ask a new cold-lunchbox example. For overclaiming, underline in this test and ask for one missing control.

Teaching and assessment notes

Timing totals 40 minutes: opening 2; retrieval 3; I do 8; We do 8; hinge 3; independent 10; review/Lundy 4; exit 2. Zero-minute slides share their stage allocation.

Default lesson is an evidence investigation. Do not imply that pupils performed the constructed experiments. Optional tactile demonstration is only at room temperature.

Use pathway support flexibly. The objective stays scientific reasoning about thermal transfer; reading, writing and social participation can be adjusted independently.

Diagrams are selective models: arrows represent net energy transfer, not a substance called cold. The ice drawing omits other energy pathways, and the bottle model simplifies thickness, fit, air movement and radiation.

Cover script: read the two temperature values first; say which is warmer; trace energy's direction; explain what changes the rate; then compare measured outcomes. Each step is one question before giving the worked explanation.

Access and responsive teaching

Supported

Use page 2 with arrow choices, the two most relevant table rows and a sentence frame. Adult reads the same science aloud. Pointing, drawing or dictated answers are valid.

Standard

Use page 3 to calculate temperature rises and justify a design decision using comparative evidence.

Stretch

Use page 4 to evaluate an unfair comparison and transfer the explanation to a warm container. Same thermal objective, deeper reasoning.

Regulation

Offer silent think time, a seated evidence role, optional partner talk and a short pause. No speed contest or public reading requirement. Pupils may change route after the hinge.

Misconceptions to check

Metal feels colder, so its temperature must be lower.

Metal and wood can have the same temperature. A warmer hand loses energy faster to the better conductor.

Check: Ask the pupil to compare both 20 °C labels before explaining the different feeling.

An insulator makes things warm.

Insulation slows energy transfer. It can slow warming of a cold object as well as cooling of a warm object.

Check: Ask why the foam-jacketed cold bottle warms less.

Cold moves out of the ice.

Net energy transfer is from warmer surroundings towards colder ice.

Check: Draw an arrow into the ice and name the warmer object.

One result proves this material is always best.

The result supports a conclusion about these jackets under these conditions. Thickness, fit and other conditions also matter.

Check: Ask which feature would need controlling before comparing materials themselves.

Word	Meaning
thermal	Related to heating and temperature.
conduction	Energy transfer through a material, or between touching materials, because of a temperature difference.
thermal conductor	A material that transfers energy by conduction readily.
thermal insulator	A material that transfers energy by conduction less readily.
evidence	Observations or measurements that help us judge a claim.

Scientific references

US Department of Energy: Heat Flow

Checked 9 September 2026. Direction of conduction and distinction between transfer mechanisms; diagrams and tasks are original.

NSW Government: Insulation

Checked 9 September 2026. Insulation reduces heat gain and heat loss; supports the cold-container transfer example. No source numerical data copied.